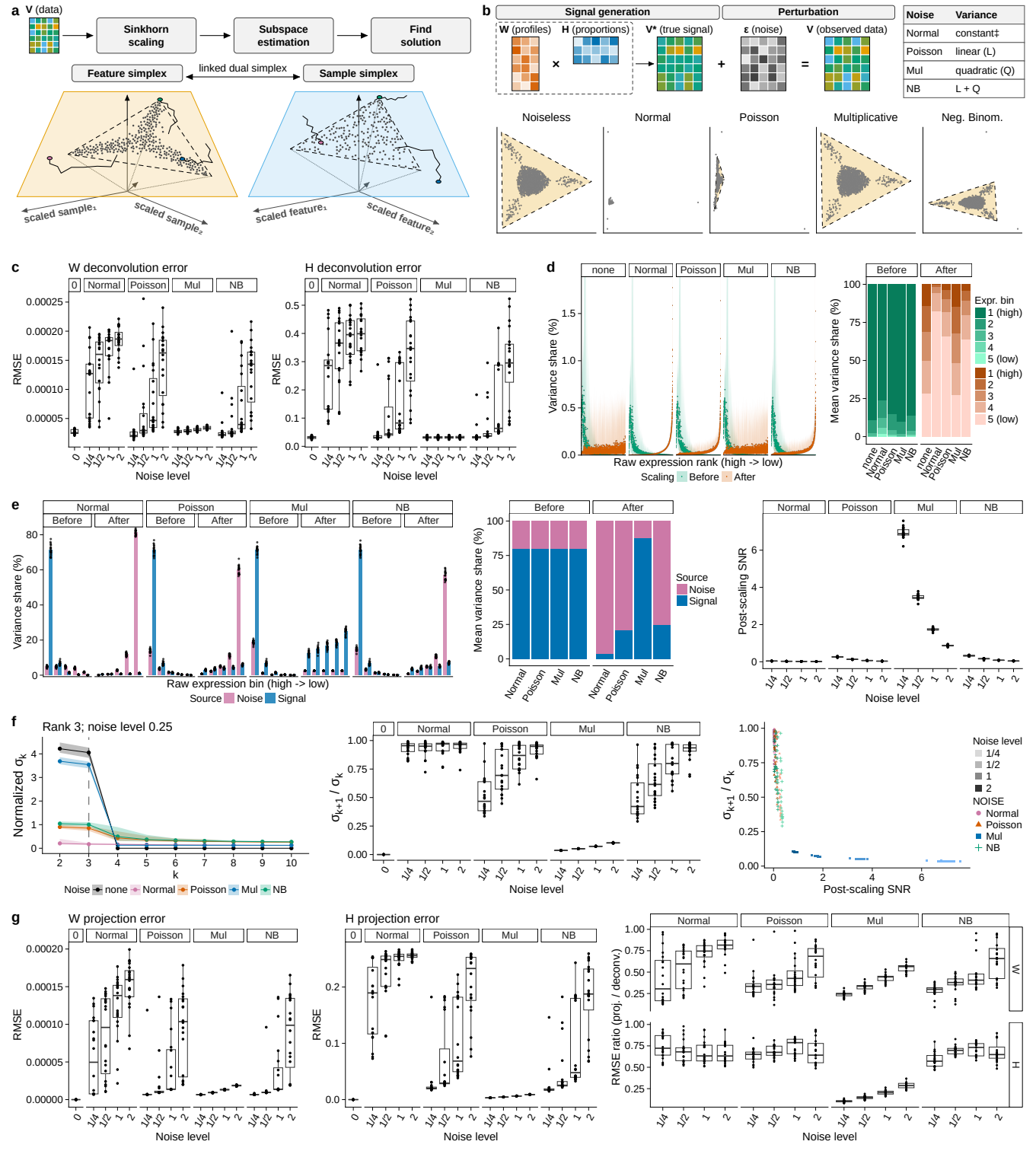

a
V (data)
Sinkhorn scaling
Subspace estimation
Find solution
Feature simplex
linked dual simplex
Sample simplex
scaled sample₁
scaled sample₂
scaled feature₁
scaled feature₂
b
Signal generation
Perturbation
W (profiles)
H (proportions)
V* (true signal)
ε (noise)
V (observed data)
Noise | Variance
Normal | constant‡
Poisson | linear (L)
Mul | quadratic (Q)
NB | L + Q
Noiseless
Normal
Poisson
Multiplicative
Neg. Binom.
c
W deconvolution error
H deconvolution error
RMSE
Noise level
d
Variance share (%)
Raw expression rank (high -> low)
Scaling Before After
Mean variance share (%)
Expr. bin
e
Variance share (%)
Raw expression bin (high -> low)
Source Noise Signal
Post-scaling SNR
f
Rank 3; noise level 0.25
Normalized σk
k
Noise none Normal Poisson Mul NB
σk+1 / σk
Noise level
Post-scaling SNR
NOISE
g
W projection error
H projection error
RMSE
RMSE ratio (proj / deconv.)